

OPEN-SOURCE AND FREE OPTIONS FOR SPATIAL DATA COLLECTION ON BIODIVERSITY

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WHAT THIS SESSION COVERS

- **The lifecycle** – where a record goes between a wet notebook and a GBIF download, and which stage each tool actually serves.
- **The options** – form-based apps, GIS-native field mapping, observation platforms, and the specialised tools for patrols and camera traps.
- **The comparison** – eight tools against ten criteria, every price and licence checked this month.
- **The trade-offs** – what the paid products genuinely do better, and what “free” costs you instead.
- **The decision** – a flowchart, and the field pitfalls that cost a season.

Written from the position of an agency standing up its own collection system, not from a vendor's. Every tool here gets a strength and a limitation, and the open ones are not assumed to win.



WHY THIS MATTERS

A record that arrives wrong is not cheaper to fix in the office. In practice it is not fixed at all – it is quietly dropped, or worse, used.



PAPER TO DIGITAL, AND WHAT IT ACTUALLY FIXES

What going digital fixes

- The **second transcription disappears**. Paper to spreadsheet is where most errors enter and none are caught.
- **Position is measured, not described**. “300 m NE of the bridge” cannot be reprojected, buffered or joined.
- **Validation moves to the moment of capture** – the one moment the observer can still walk back and look.
- **Metadata is written while it is true**: observer, method, effort, accuracy.

What it does not fix

- A **weak protocol collected faster** is still a weak protocol, now with more rows.
- **Taxonomy** still needs a backbone and a reviewer.
- **Absence and effort** are recorded only if the form insists on them.
- A phone in a wet pocket at minus five degrees is **less reliable than a pencil**. Plan for both.



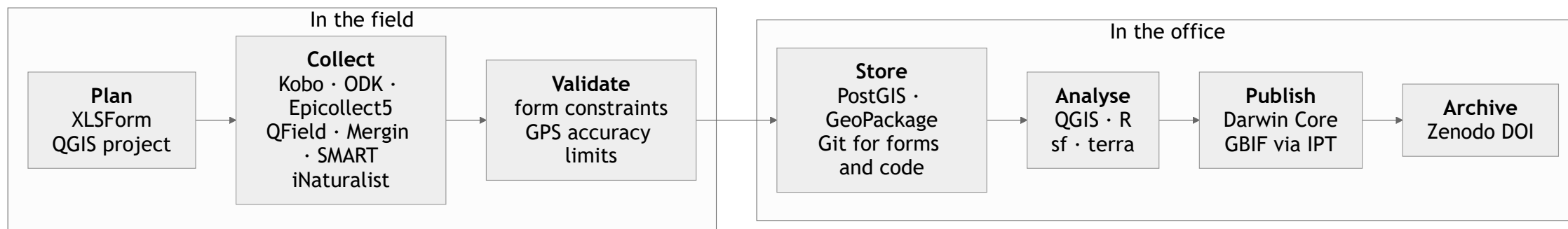
THE LIFECYCLE A RECORD TRAVELS

- **Plan** – the protocol, the target features, who collects, what counts as effort. Decided before the form is built.
- **Collect** – the form or the map on the device, offline.
- **Validate** – constraints at capture, review afterwards, taxon matching.
- **Store** – one authoritative copy, versioned and backed up.
- **Analyse** – scripted and re-runnable, not a spreadsheet lineage.
- **Publish and archive** – Darwin Core to GBIF, a DOI that still resolves in ten years.

FAIR – findable, accessible, interoperable, reusable – is the test at the end ([Wilkinson et al. 2016](#)). The practical form of it: could a colleague re-run this in five years without telephoning you?



THE LANDSCAPE, MAPPED ONTO THE LIFECYCLE



COLLECTING IN THE FIELD

Four families of tool, and the choice between them is made by the geometry and the crew, not by the price list.



FORM-BASED COLLECTION: THE OPEN OPTIONS

 KoboToolbox · AGPL-3.0

- **EU server in Ireland**, a global server, or self-hosted.
- Free for non-profits and government: **5 000 submissions a month, 1 GB**.
- Limitation: **those caps**; the next tier is \$159 a month.

 **ODK Collect + Central** · Apache-2.0

- The project Kobo forks. **Self-hosted it costs only the server and the admin.**
- Limitation: **ODK Cloud from \$199 a month**, so self-host Central.

 epicollect5 · MIT server

- Free with **no stated limits**, run by the Centre for Genomic Pathogen Surveillance at Oxford. Nothing to set up.
- Limitation: **points only**, no lines or polygons, and the API is read-only. Offered “as-is” by a small research team.

All three share

- **XLSForm or a web builder**, offline Android collection, and a defined export.
- Geometry is stored in **WGS84 only**. Reprojection is your job, downstream.

WHAT A FORM ENFORCES AT THE POINT OF CAPTURE

An XLSForm survey sheet is a spreadsheet. These four rows are most of what separates a clean dataset from a dirty one.

type	name	label	constraint / relevant
geopoint	pond_loc	Pond centre	<code>selected-at(\${pond_loc}, 3) < 15</code>
select_one taxon	species	Species observed	from a controlled list
integer	count	Individuals counted	<code>. >= 0 and . <= 5000</code>
begin_repeat	obs	One per species	relevant: <code>\${survey_done} = 'yes'</code>

The first constraint refuses a fix worse than 15 m accuracy. The third catches a typed extra zero. The fourth is how one visit yields several occurrence rows instead of one wide row you have to unpick later ([XLSForm, n.d.](#); [ODK, n.d.-a](#)).

FORM-BASED COLLECTION: THE PAID COMPARATORS



ArcGIS Survey123 · proprietary

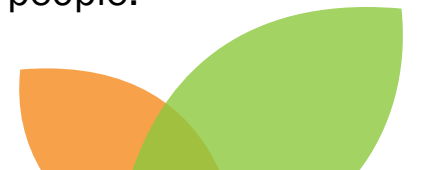
- **XLSForm plus Esri extensions**, so a form is portable in, though not cleanly out.
- Publishes straight into the organisation's existing feature services and dashboards.
- Limitation: **no price on the product page**; it comes with ArcGIS user types and consumes credits for storage and reports.



Fulcrum · proprietary

- **\$43–55 per user per month**, minimum five users, billed annually.
- Polished builder, strong offline behaviour, device management, real support.
- Limitation: **proprietary end to end**, and the hosting location is not published – awkward if data residency is a legal requirement.

Both are worth their money in the right organisation. Neither is worth it to stand up one survey of twelve people.



GIS-NATIVE FIELD MAPPING

QField + QFieldCloud · GPL-2.0+ / MIT

- Your **QGIS project itself** goes to the device: layers, styling, snapping, attribute forms, offline basemaps.
- Cloud is **free up to 100 MB**, then €18 per user per month; private instances on request, hosted in Switzerland.
- Limitation: **it presumes QGIS skills**. Someone must own the project file.

merginmaps · AGPL-3.0 / GPL-2.0

- Same idea, with **versioned sync** as the centrepiece and a Community Edition you can host yourself at no licence cost.
- Cloud from **€15.80 per contributor per month**; data residency only on the Enterprise plan.
- Limitation: same QGIS dependency, and the free cloud tiers are narrow – 500 MB for small non-profits.



WHEN THE MAP IS THE WORKING DOCUMENT

Use GIS-native when

- The feature is a **polygon or a line** – a habitat patch, a transect, a management compartment.
- The surveyor must **see existing features** and edit them, not just add new points.
- **Topology matters**: adjoining patches, no gaps, no overlaps.
- The work happens in a **national CRS** and must stay there.

Use a form when

- Records are **independent point observations**.
- The crew is **large, changing, or minimally trained**.
- The survey is **short-lived** and must be running next week.

 **ArcGIS Field Maps** is the paid equivalent: excellent offline map areas and device management, no XLSForm, and again no price on the product page.

OBSERVATION PLATFORMS AND CITIZEN SCIENCE

- **iNaturalist** – identification support, a large community, research-grade records flow weekly to GBIF. Limitation: you are a guest on someone else's platform and you do not set the licence.
- **Observation.org** – strong in Europe, validation by regional experts, a species-rich app. Limitation: national portals vary in how they moderate.
- **eBird** – the deepest structured effort data in existence, for birds only.
- **National recording systems** – the record stays in national custody, which is usually the point.

These are a **complement to institutional monitoring, not a substitute**. Volunteer effort is uneven in space and time; a designed survey is not.



WHO OWNS THE RECORD, AND WHAT THE LICENCE DECIDES

Platform	Licence on GBIF	Records in GBIF
iNaturalist research-grade	CC BY-NC 4.0	166.9 million
Observation.org	CC BY-NC 4.0	135.1 million
eBird (EOD)	CC BY 4.0	1 775.8 million

- **GBIF accepts only CC0, CC BY and CC BY-NC.** Anything else never leaves the platform.
- **NC is not free.** It bars uses a consultant or a developer may need, and the boundary is untested.
- The **contributor**, not the institution, chooses the licence on these platforms.



SPECIALISED TOOLS, ONE SLIDE

- **SMART** – patrol and law-enforcement monitoring for protected areas. Free and open, maintained by an eight-organisation partnership. Limitation: it is a management system, not a biodiversity database.
- **CyberTracker** – icon-driven, works for non-literate observers. Free for Indigenous, citizen-science, humanitarian and student projects; otherwise ZAR 350–3 400 a year by data volume.
- **Camera traps and acoustics** – **Agouti** (Wageningen and INBO, data stays yours) and **Wildlife Insights** (free for many users, but **government bodies are expected to pay** unless in a transitioning economy).
- **Camtrap DP** is the open exchange format maintained under TDWG. Agouti exports it; GBIF ingests it via the IPT ([Bubnicki et al. 2024](#)).



FROM THE DEVICE TO THE DATABASE

The gap where most systems leak: the manual CSV export, emailed, renamed and edited by hand.



PULL THE DATA, DO NOT EXPORT IT

Why the manual export fails

- It is **undated, unversioned and unattributable** the moment it is downloaded.
- Someone always **opens it in Excel**, and the dates and leading zeros change.
- Corrections made on the server **never reach the copy** already in circulation.
- It cannot be **re-run**, so the analysis is not reproducible.

The scripted alternative

```
library(robottoolbox)
kobo_setup(
  url = "https://eu.kobotoolbox.org",
  token = Sys.getenv("KOBO_TOKEN")
)
ponds <- kobo_asset("aBcD...") |>
  kobo_submissions()
```

One function, the live server, a token from an environment variable and never from the script. **ruODK** does the same for ODK Central; both QGIS-based tools hand you a GeoPackage that **sf::st_read()** opens directly.

VERSION THE FORM, NOT ONLY THE DATA

- A **form definition is source code**. Changing a question mid-season changes the meaning of a column, and without a history nobody can say when.
- Keep in Git: the **XLSForm or QGIS project**, the **processing scripts**, the **codebooks**, and the **documentation**.
- Tag a release when the season opens. Then **v1 data and v2 data are distinguishable** rather than merged in silence.
- GitHub Pages publishes the documentation from the same repository, so the method people read is the method that ran.

Not the raw data itself: that belongs in a database with proper access control. Git holds what defines and transforms it.



PUBLISHING AND ARCHIVING

- **Darwin Core** is the shared vocabulary. Map your fields to it once, at the design stage, not after collection.
- **GBIF IPT** publishes the dataset and makes it citable; the record keeps your institution as publisher.
- **Zenodo** gives a DOI to the snapshot the analysis actually used, which is what a paper or a report must cite.
- **Metadata is written at collection time.** Retrospective metadata is reconstruction, and it shows.

Covered properly in yesterday's GBIF session and in this morning's national portal and geoportal sessions – this slide is only the handover.

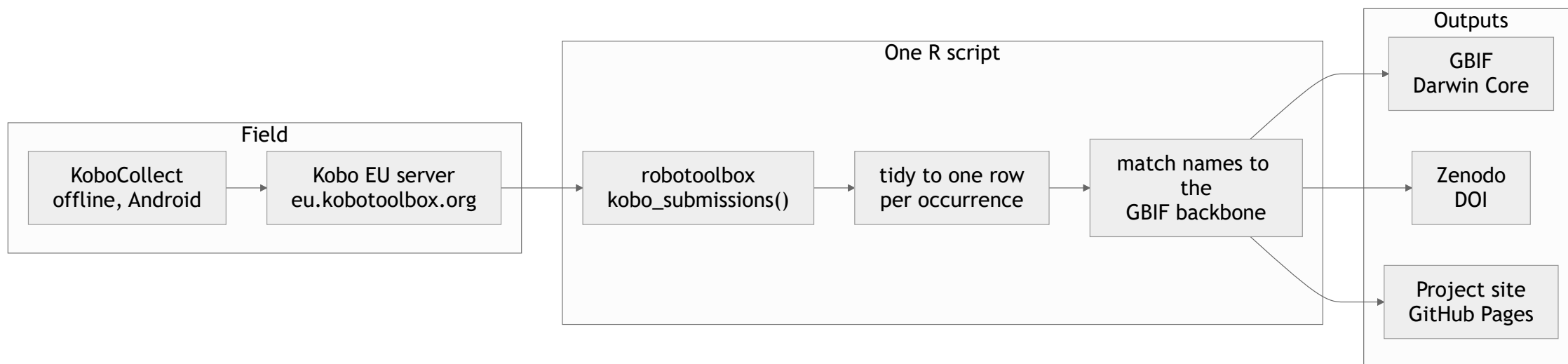


A WORKED PILOT: BIODIVPOND

Biodiversity monitoring of ponds – a Biodiversa+ pilot running 2026 to 2028, testing whether novel methods survive contact with real field crews.



BIODIVPOND: THE PIPELINE



BIODIVPOND: WHAT WORKED, WHAT HURT

What the design got right

- **The EU server was chosen before the first form**, not argued about afterwards.
- **One row per species occurrence** from the start, so nothing had to be unpivoted later.
- **Names matched to the GBIF backbone inside the pipeline**, so a failure is visible on every run rather than at publication.

What was painful

- **Versioning the form.** Once submissions arrive, every change is a new version, and each row has to be traced to the version that produced it.
- **Building it under time pressure.** The field season does not move, so the form was finished against the deadline rather than ahead of it.
- **Too little testing.** What a dry run would have caught was found in the field instead.

One lesson under all three: a form is software, and it needs a release cycle – build, test, freeze, version.



CHOOSING

Eight tools, ten criteria. Prices, licences and hosting checked on one day and already ageing.



HEAD TO HEAD: COST, EFFORT AND CONTROL

Tool	Cost model	Licence	Hosting and sovereignty	Learning curve	Lock-in
KoboToolbox	Free for non-profits and government: 5 000 subs/month, 1 GB; then \$159/month	AGPL-3.0	EU server (Ireland) or global; self-hostable	Medium (XLSForm)	Low: open source, full export
ODK Collect + Central	Self-hosting free; ODK Cloud from \$199/month	Apache-2.0	Self-host anywhere; Cloud in EU, US or India	Medium; server admin	Low: open source, self-hosted
Epicollect5	Free, no stated limits (fair use)	MIT (server)	Oxford (CGPS); self-host possible	Low	Low, but one small team
QField + QFieldCloud	QField free; cloud free 100 MB, then EUR 18/user/month	GPL-2.0+ / MIT	Swiss datacentres; self-host the MIT server	High: needs QGIS skills	Low: QGIS project + GeoPackage
Mergin Maps	Community Edition free self-hosted; cloud EUR 15.80/contributor/month, billed annually	AGPL-3.0 / GPL-2.0	Cloud; data residency on Enterprise; self-host CE	High: needs QGIS skills	Low: open CE, GeoPackage
ArcGIS Survey123	No price on the product page; comes with ArcGIS user types	Proprietary	Esri cloud, or ArcGIS Enterprise on-premises	Low if already on ArcGIS	High: tied to ArcGIS licensing
ArcGIS Field Maps	No price on the product page; comes with ArcGIS user types	Proprietary	Esri cloud, or ArcGIS Enterprise on-premises	Low if already on ArcGIS	High: tied to ArcGIS licensing
Fulcrum	USD 43-55/user/month; minimum 5 users, billed annually	Proprietary	Vendor cloud (AWS); region not published	Low	Medium: proprietary platform

As checked on 23 September 2026. Source URLs per row are in [data/comparison.csv](#); re-check before you quote any of it.

HEAD TO HEAD: FIELDWORK AND INTEGRATION

Tool	Offline	Geometry	CRS	Form logic	API and R
KoboToolbox	Yes; MBTiles basemaps	Point, line, polygon	WGS84 only	XLSForm: full	REST; R robotoolbox
ODK Collect + Central	Yes; MBTiles basemaps	Point, line, polygon	WGS84 only	XLSForm: full	OData + REST; R ruODK
Epicollect5	Yes; no offline basemaps	Point only	WGS84 only	Branch + group; limited	Read-only REST
QField + QFieldCloud	Yes; whole QGIS project offline	Point, line, polygon	Any CRS QGIS supports	QGIS forms + constraints	GeoPackage; R sf, PyQGIS
Mergin Maps	Yes; whole QGIS project offline	Point, line, polygon	Any CRS QGIS supports	QGIS forms + constraints	GeoPackage; R sf, Python client
ArcGIS Survey123	Yes; offline basemap areas	Point, line, polygon	Any CRS via ArcGIS	XLSForm + Esri extensions	ArcGIS REST; R arcgislayers
ArcGIS Field Maps	Yes; offline map areas	Point, line, polygon	Any CRS via ArcGIS	Smart forms; no XLSForm	ArcGIS REST; R arcgislayers
Fulcrum	Yes; offline basemaps	Point, line, polygon	WGS84 only	Drag-and-drop + conditions	REST API (Elite tier and up)

As checked on 23 September 2026. The **crs** column is the one that decides more projects than the price column does.

WHAT THE PAID TOOLS ACTUALLY BUY YOU

- **Support with a deadline attached.** An open-source forum answers when someone feels like it. A contract answers on Tuesday.
- **Device management** – enrolling, wiping and updating fifty tablets without touching them.
- **A polished builder** that a non-technical colleague can use unaided.
- **Enterprise integration** – single sign-on, existing feature services, dashboards your director already reads.
- **Someone to blame.** In a public body that is a real procurement argument, not a joke.



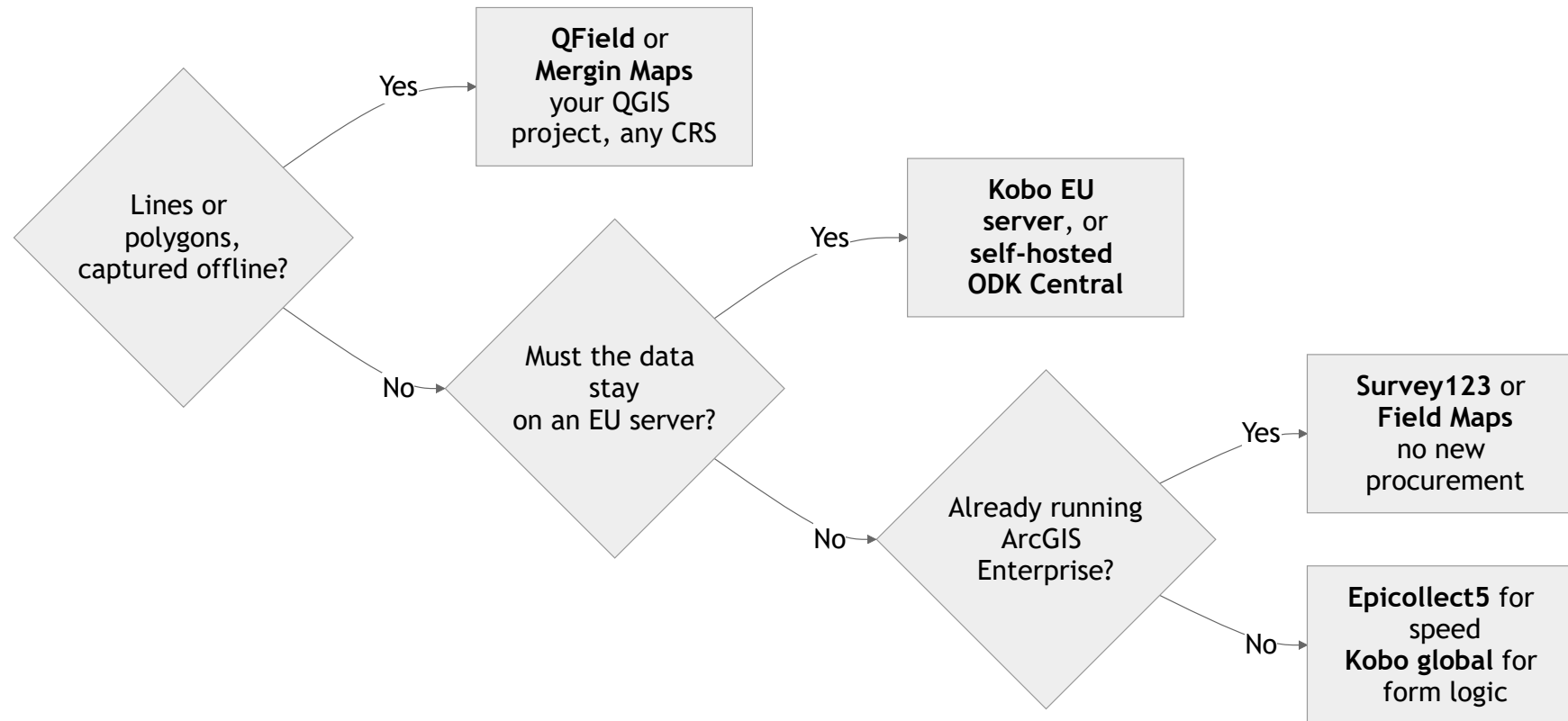
THE HIDDEN COSTS OF FREE

- **Self-hosting is a post, not a download.** ODK Central needs a server, backups, TLS certificates and somebody who notices when it stops.
- **Donor-funded servers change terms.** Kobo's free tier was re-cut in 2023, to 5 000 submissions a month and 1 GB.
- **Training is the largest line either way,** and it does not shrink because the licence is zero.
- **"As-is" means as-is.** Epicollect5 is run by a small research team, and says so plainly.
- **Migration you postpone is migration you pay for,** with interest.

Open source removes the licence fee and the lock-in. It does not remove the staff cost, and pretending otherwise is how these projects fail in year two.



A DECISION GUIDE



FIELD PITFALLS THAT COST A SEASON

- **Record GPS accuracy as a field**, and refuse fixes worse than your threshold. A point without an accuracy value cannot be assessed later.
- **Fix the CRS before the first survey.** Kosovo's national system is KOSOVAREF01 / Balkans zone 7, **EPSG:9141**; devices give you WGS84 (EPSG:4326). Decide where the conversion happens, once.
- **Generalise sensitive species before publication**, not before storage. Keep the precise record internally.
- **Charge, spare and test the devices.** Cold kills batteries faster than the survey does.
- **Freeze the form when the season opens.** If it must change, version it and record which version produced which row.
- **Back up off the device daily**, and test a restore before you need one.
- **Write metadata at capture.** Anything written in November is memory, not metadata.

WHERE TO START ON MONDAY

- **Pick one survey**, not a system. Build its form, run it with four people for a week, and count what broke.
- **Take a starter XLSForm** with geopoint, an accuracy constraint, a controlled taxon list and a repeat group. Both Kobo and ODK will load it.
- **Or take a starter QGIS project for Annex I habitat mapping** – segment layer, habitat code list, share and condition fields, one offline orthophoto – and open it in QField.
- **Put both under version control the same day.**
- **Decide the publication licence before collection**, because it constrains the consent text you need.

All of it, including this deck and the comparison data: github.com/jonasgaigr/kosovo-biodiversity-data-workshop-2026



SOURCES

Bubnicki, Jakub W., Ben Norton, Steven J. Baskauf, et al. 2024. 'Camtrap DP: An Open Standard for the FAIR Exchange and Archiving of Camera Trap Data'. *Remote Sensing in Ecology and Conservation* 10 (3): 283–95. <https://doi.org/10.1002/rse2.374>.

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The full source list, with an access date for every figure in this deck, is [references.bib](#); the per-row sources behind the comparison are the [source](#) and [checked_on](#) columns of [data/comparison.csv](#). Product names and logos are the trademarks of their respective owners, used here to identify the products compared; no affiliation or endorsement is implied. ODK is free and open-source software — getodk.org.

THANK YOU FOR YOUR ATTENTION!

github.com/jonasgaigr/kosovo-biodiversity-data-workshop-2026



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